

TUGAS JARKOM PRAKTIKUM WEEK 12

ICMP

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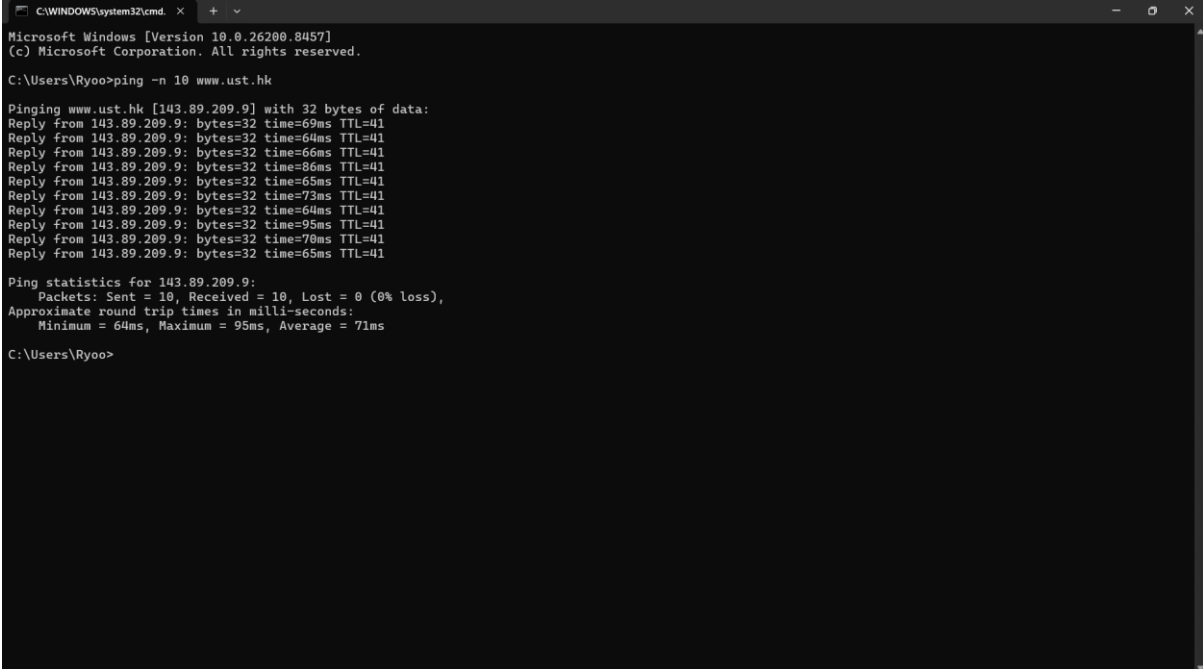
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Tujuan :

1. Mahasiswa dapat menginvestigasi cara kerja protokol ICMP menggunakan Wireshark
2. Mahasiswa dapat membuat program ICMP Pinger

ICMP dan PING

1. ping -n 10 www.ust.hk



```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26280.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Ryoo>ping -n 10 www.ust.hk

Pinging www.ust.hk [143.89.209.9] with 32 bytes of data:
Reply from 143.89.209.9: bytes=32 time=69ms TTL=41
Reply from 143.89.209.9: bytes=32 time=64ms TTL=41
Reply from 143.89.209.9: bytes=32 time=66ms TTL=41
Reply from 143.89.209.9: bytes=32 time=86ms TTL=41
Reply from 143.89.209.9: bytes=32 time=65ms TTL=41
Reply from 143.89.209.9: bytes=32 time=73ms TTL=41
Reply from 143.89.209.9: bytes=32 time=64ms TTL=41
Reply from 143.89.209.9: bytes=32 time=95ms TTL=41
Reply from 143.89.209.9: bytes=32 time=79ms TTL=41
Reply from 143.89.209.9: bytes=32 time=65ms TTL=41

Ping statistics for 143.89.209.9:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 64ms, Maximum = 95ms, Average = 71ms

C:\Users\Ryoo>
```

2. ICMP (Echo Ping Request)

The screenshot shows a Wireshark capture of ICMP Echo (ping) requests. The packet list on the left shows a series of requests from 192.168.0.102 to 143.89.209.9. The selected packet (No. 260) is an Echo (ping) request. The packet details pane shows the following fields:

- Internet Control Message Protocol
- Type: Echo (ping) request (8)
- Code: 0
- Checksum: 0x4d59 [correct]
- [Checksum Status: Good]
- Identifier (BE): 1 (0x0001)
- Identifier (LE): 256 (0x0100)
- Sequence Number (BE): 2 (0x0002)
- Sequence Number (LE): 512 (0x0200)
- [Response frame: 261]
- Data (32 bytes)

The packet bytes pane shows the raw data of the request, including the ICMP header and the 32-byte data field.

3. ICMP (Echo Ping Reply)

The screenshot shows a Wireshark capture of ICMP Echo (ping) replies. The packet list on the left shows a series of replies from 143.89.209.9 to 192.168.0.102. The selected packet (No. 261) is an Echo (ping) reply. The packet details pane shows the following fields:

- Internet Control Message Protocol
- Type: Echo (ping) reply (0)
- Code: 0
- Checksum: 0x5559 [correct]
- [Checksum Status: Good]
- Identifier (BE): 1 (0x0001)
- Identifier (LE): 256 (0x0100)
- Sequence Number (BE): 2 (0x0002)
- Sequence Number (LE): 512 (0x0200)
- [Request frame: 260]
- [Response time: 69,727 ms]
- Data (32 bytes)

The packet bytes pane shows the raw data of the reply, including the ICMP header and the 32-byte data field.

ICMP dan Traceroute

1. c:\windows\system32\tracert www.inria.fr

```
C:\WINDOWS\system32\cmd. x + v
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Ryoo>c:\windows\system32\tracert www.inria.fr

Tracing route to inria.fr [128.93.162.83]
over a maximum of 30 hops:
  0  33 ms  42 ms  6 ms  192.168.0.1
  1  18 ms  6 ms  4 ms  192.168.1.1
  2  24 ms  13 ms  21 ms  10.114.0.1
  3  26 ms  8 ms  5 ms  188.250.252.69
  4  *  41 ms  *  188.240.190.77
  5  53 ms  58 ms  *  188.240.190.77
  6  187 ms  179 ms  180 ms  188.240.192.233
  7  198 ms  193 ms  193 ms  188.240.196.1
  8  273 ms  297 ms  290 ms  renater.par.franceix.net [37.49.236.19]
  9  192 ms  192 ms  196 ms  hu0-4-0-1-ren-nt-orsay-rtr-091.noc.renater.fr [193.51.180.43]
 10  229 ms  209 ms  198 ms  inria-roquencourt-vl1631-te1-4-inria-rtr-021.noc.renater.fr [193.51.184.177]
 11  291 ms  283 ms  287 ms  unit240-reth1-vfw-ext-dcl.inria.fr [192.93.122.19]
 12  193 ms  197 ms  193 ms  prod-inriafr-cms.inria.fr [128.93.162.83]

Trace complete.

C:\Users\Ryoo>h
```

2. ICMP after c:\windows\system32\tracert www.inria.fr in CMD

```
*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.      icmp      Source      Destination  Protocol Length Info
-----
85 12.2340709... 192.168.0.102 128.93.162.83 ICMP 106 Echo (ping) request id=0x0001, seq=51/13056, ttl=1 (no response found!)
86 12.2445549... 192.168.0.1 192.168.0.102 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
87 12.2452789... 192.168.0.102 128.93.162.83 ICMP 106 Echo (ping) request id=0x0001, seq=52/13312, ttl=1 (no response found!)
88 12.2464242... 192.168.0.1 192.168.0.102 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
89 12.2468882... 192.168.0.102 128.93.162.83 ICMP 106 Echo (ping) request id=0x0001, seq=53/13568, ttl=1 (no response found!)
90 12.2485626... 192.168.0.1 192.168.0.102 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
100 12.6682478... 192.168.0.1 192.168.0.102 ICMP 120 Destination unreachable (Port unreachable)
106 14.1807911... 192.168.0.1 192.168.0.102 ICMP 120 Destination unreachable (Port unreachable)
123 15.7009131... 192.168.0.1 192.168.0.102 ICMP 120 Destination unreachable (Port unreachable)
138 18.2256972... 192.168.0.102 128.93.162.83 ICMP 106 Echo (ping) request id=0x0001, seq=54/13824, ttl=2 (no response found!)
139 18.2401958... 192.168.1.1 192.168.0.102 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
140 18.2477031... 192.168.0.102 128.93.162.83 ICMP 106 Echo (ping) request id=0x0001, seq=55/14080, ttl=2 (no response found!)
142 18.2530041... 192.168.1.1 192.168.0.102 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
143 18.2533691... 192.168.0.102 128.93.162.83 ICMP 106 Echo (ping) request id=0x0001, seq=56/14336, ttl=2 (no response found!)
144 18.2579009... 192.168.1.1 192.168.0.102 ICMP 134 Time-to-live exceeded (Time to live exceeded in transit)
148 18.2756217... 192.168.1.1 192.168.0.102 ICMP 120 Destination unreachable (Port unreachable)
151 19.7698834... 192.168.1.1 192.168.0.102 ICMP 120 Destination unreachable (Port unreachable)
157 21.2844177... 192.168.1.1 192.168.0.102 ICMP 120 Destination unreachable (Port unreachable)

Frame 85: Packet, 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on interface
Ethernet II, Src: LiteonTechno af:cf:7d (24:b2:b9:af:cf:7d), Dst: TPLink 5b:1b:b4 (bc:07:11:5b:1b:b4)
Internet Protocol Version 4, Src: 192.168.0.102, Dst: 128.93.162.83
Internet Control Message Protocol

0000  bc 07 1d 5b 1b b4 24 b2  b9 af cf 7d 08 00 45 00  ...[. $ ...]--E
0010  00 5c 6b 5c 00 00 01 01  6a 86 c0 a8 00 66 80 5d   \K\...j...f.]
0020  a2 53 00 00 7f cb 00 01  00 33 00 00 00 00 00 00   $.....3.....
0030  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   .....
0040  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   .....
0050  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   .....
0060  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   .....

Packets: 669 - Displayed: 88 (13.2%) - Dropped: 0 (0.0%) Profile: Default
```

3. Detail Time-To-Live Exceeded

The screenshot displays a Wireshark packet capture of ICMP Echo (ping) requests. The packet list shows frames 85 through 157, with the selected frame 86 expanded to show the ICMP Echo (ping) request details. The packet details pane shows the following information:

- Frame 86: Packet, 134 bytes on wire (1072 bits), 134 bytes captured (1072 bits) on interface 0
- Ethernet II, Src: TP-Link 5b:1b:b4 (bc:07:1d:5b:1b:b4), Dst: LiteonTechno_af:cf:7d (24:b2:b0:00:00:00)
- Internet Protocol Version 4, Src: 192.168.0.1, Dst: 128.93.162.83
- Internet Control Message Protocol
 - Type: Time-to-live exceeded (11)
 - Code: 0 (Time to live exceeded in transit)
 - Checksum: 0xf4ff [correct]
 - [Checksum Status: Good]
 - Unused: 00000000
 - Internet Protocol Version 4, Src: 192.168.0.102, Dst: 128.93.162.83
 - Internet Control Message Protocol

The packet bytes pane shows the raw data of the packet, including the Ethernet II header, Internet Protocol Version 4 header, and Internet Control Message Protocol header.